

PRICING & TECHNOLOGY BROCHURE

AI Kannada Tele-Caller

Bengaluru "Kanglish" voice agent

A production-grade outbound & inbound phone agent that speaks natural, code-mixed Bengaluru Kannada — built on the **Sarvam × Gemini × Exotel** stack. This brochure documents the implemented architecture and the exact, component-level cost to run it.

All-in cost ≈ ₹2.3–2.5 / min (~\$0.027)

Powered by Sarvam Saarika STT · Google Gemini · Exotel Voice · Pipecat orchestration

1 What it is & how it works

artintele.ai is a real-time voice tele-caller that holds a natural phone conversation in casual Bengaluru **Kanglish** (spoken Kannada mixed with English). It handles outbound campaigns and inbound calls, understands the caller by ear, and answers in a warm, human, local-sounding voice at roughly **1.2–1.7 s** perceived latency per turn. The system is orchestrated with **Pipecat 0.0.108** and runs one of three interchangeable voice engines.

The deployed pipeline (“hybrid” engine)



The caller's audio arrives over **Exotel**; **Sarvam Saarika** transcribes the code-mixed speech; **Google Gemini** (a Flash-Lite text model) generates a short reply constrained to the Bengaluru dialect by an in-context system prompt + slang lexicon; and **Gemini Cloud TTS** (voice “Aoede”) speaks it back. Two latency systems run alongside: a **predictive answer cache** that pre-writes likely answers while the caller is still talking, and a **tiered filler injector** that plays a natural “one minute sir...” hold if a turn runs slow.

Implemented components

Layer	Technology (as deployed)	Role
Telephony	Exotel Voice + Media Streams	Carries the call; streams 8 kHz audio to the app
Speech-to-text	Sarvam Saarika v2.5 (kn-IN)	Transcribes code-mixed Kannada/English
Language brain	Google Gemini 3.1 Flash-Lite	Generates the dialect-locked reply (≤ 160 tokens)
Voice / TTS	Gemini 2.5 Flash TTS — voice “Aoede”	Speaks the reply (24 kHz, downsampled to phone)
Anticipation	Gemini Flash-Lite + gemini-embedding	Predictive cache: pre-answers likely next questions
Orchestration	Pipecat 0.0.108 + Silero VAD	Turn-taking, barge-in, service glue, fillers
App host	FastAPI + Uvicorn	Webhook, media websocket, live dashboard

Alternate engines (same codebase, one toggle): **gemini-direct** — Gemini Live native audio hears and speaks Kannada directly (no STT/TTS stage; lowest latency, premium voice); and **sarvam** — Gemini text + Sarvam Bulbul TTS (India-native speaker, lowest voice cost). Pricing for all three is compared in section 4.

2 Verified rate card

Every meter the system touches, at published rates (Jul 2026). Nothing here is estimated — these are list prices; the per-minute math in section 3 is built from them.

Component	Provider / model	List rate	Meter
Speech-to-text	Sarvam Saarika v2.5	₹30 / hour (= ₹0.50/min)	Per second of audio
Text-to-speech (deployed)	Gemini 2.5 Flash TTS	\$0.50/1M in · \$10.00/1M out	Text in + audio out*
Text-to-speech (alt)	Sarvam Bulbul v2 / v3	₹15 / ₹30 per 10K chars	Per character
Language model	Gemini 3.1 Flash-Lite	\$0.10/1M in · \$0.40/1M out	Per token
Embeddings (cache)	gemini-embedding	\$0.15 / 1M in	Per token
Live native audio (alt)	Gemini 2.5 Flash Native Audio	\$3/1M in · \$12/1M out	Per audio token
Telephony	Exotel outbound (India)	₹0.80–1.00/min + 18% GST	Per minute**

* Gemini audio tokenises at a fixed **25 tokens / second** of speech (1M output tokens ≈ 11.1 hours of audio). ** Exotel does not publish a fixed public per-minute rate; it bills against prepaid credit bundles and volume/contract terms. ₹0.80–1.00/min + GST is the working figure used here for outbound calls to Indian mobiles — confirm your rate with Exotel for exact billing.

Why this stack is priced the way it is

- **Sarvam** is billed in INR and is dramatically cheaper for Indian-language STT than global clouds — STT is effectively a rounding error at ₹0.50/min.
- **Gemini Flash-Lite** is the cheapest capable text tier; the reply is hard-capped at 160 tokens, so LLM spend stays tiny even with the large dialect prompt.
- **Voice (TTS) is the largest AI line-item** because audio output tokens are the premium meter. The bot only speaks ~half the minute, which keeps it in check.
- **Exotel** is chosen over Twilio specifically for India: it roughly halves the telephony leg (Twilio-to-India ran ≈₹9–13/min).

3 Per-minute cost — deployed hybrid engine

Modelled on a realistic **balanced minute**: the caller speaks ~30 s and the bot speaks ~30 s across ~4 conversational turns, with the ~5K-token dialect prompt in context. FX \$1 ≈ ₹86.

Line item	What is metered	Tokens / units	Cost (₹/min)
Sarvam STT	~30 s of caller audio	0.5 min @ ₹0.50/min	₹0.25
Gemini LLM (brain)	~4 turns × (prompt+context in, ≤160 out)	~24K in · ~0.4K out	₹0.22
Predictive cache	pre-gen + embeddings each turn	~10K in · ~1.6K out	₹0.14
Gemini TTS (voice)	~30 s of bot speech @ 25 tok/s	~750 audio tok out	₹0.66
AI subtotal	Sarvam + Gemini only		₹1.27
Exotel telephony	1 outbound min + 18% GST	₹0.80–1.00 + GST	₹0.94–1.18
ALL-IN TOTAL	per conversational minute	~\$0.026–0.029	₹2.2–2.5

PER MINUTE

₹2.3

all-in, typical call

AI ONLY

₹1.27

Sarvam + Gemini stack

VOICE SHARE

~52%

TTS is the top AI cost

Reading this

Telephony (Exotel) and the AI stack each carry roughly **half** the per-minute cost. Within the AI half, **voice synthesis dominates** — the STT, the LLM brain, and the entire predictive-cache machinery together cost less than the TTS. That is the key optimisation lever (section 5): anything that shortens bot speech, or swaps to a cheaper voice, moves the needle most.

4 Engine options compared

The same codebase runs three voice engines behind one VOICE_ENGINE toggle. All three land near ₹2.5/min for a short call — the real trade-off is **voice quality, latency, and how cost behaves on long calls**.

Engine	Pipeline	Voice / latency	₹/min (short)	Long-call risk
hybrid deployed	Sarvam STT › Gemini LLM › Gemini TTS	Warm Aoede voice; ~1.3–1.7 s	₹2.2–2.5	Low — flat
gemini-direct best voice	Gemini Live native audio (no STT/TTS)	Best Kannada; lowest ~1.0–1.2 s	₹2.1–2.3	High — ₹4–6/min on long calls*
sarvam cheapest voice	Sarvam STT › Gemini LLM › Sarvam Bulbul TTS	India-native speaker; ~1.5 s	₹2.2–2.5	Low (v2) – v3 +₹0.7

* Gemini Live reprocesses the growing conversation each turn, so cost climbs with call length even with context-window compression enabled. Best for short, high-value calls where voice quality is paramount. The deployed **hybrid** engine keeps cost flat across call length, which is why it is the default for volume campaigns.

5 Cost-optimisation levers

Lever	Effect
Keep replies short (≤160 tok cap)	Directly cuts the #1 cost (TTS audio out) and TTS latency. Already enforced.
Predictive cache hits	A cache hit skips a full LLM turn › ~1–1.5 s faster and one fewer LLM call.
Context caching on the prompt	The ~5K dialect prompt is re-sent each turn; caching it cuts LLM input cost further.
Use sarvam Bulbul v2 voice	Cheapest voice tier; comparable to Gemini TTS cost with an India-native speaker.
Stay on Exotel, not Twilio	Halves the telephony leg for India (≈₹1/min vs ≈₹9–13/min).
Keep calls focused / short	Fewer minutes = linear savings; critical for the gemini-direct engine.

6 What it costs at volume

Usage cost only, at the deployed hybrid rate of **~₹2.4/min** and a typical **3-minute** call. Sarvam, Gemini and Exotel are all pay-as-you-go, so cost scales linearly with connected minutes.

Volume (calls / month)	Connected minutes	Monthly cost (₹)	Monthly cost (USD)
1 call × 3 min	3 min	₹7.2	~\$0.08
100 calls × 3 min	300 min	₹720	~\$8
1,000 calls × 3 min	3,000 min	₹7,200	~\$84
10,000 calls × 3 min	30,000 min	₹72,000	~\$840
100,000 calls × 3 min	300,000 min	₹7,20,000	~\$8,370

Per-minute cost is **flat** on the hybrid engine regardless of volume; there is no step pricing in the model above. Committed Exotel bundles and Gemini/Sarvam volume terms can reduce these further at the higher tiers.

7 Fixed & one-time costs

Beyond per-minute usage, running the service needs a small always-on footprint. These are typical, not billed by the AI vendors:

Item	Type	Indicative cost	Notes
App host (FastAPI + Pipecat)	Monthly	₹1,500–4,000 (~\$20–50)	One small always-on VM / container
Public ingress	Monthly	₹0 – low	cloudflared tunnel (free) › named tunnel / cloud for prod
Exotel number + plan minimum	Monthly	Per Exotel plan	Exophone rental + prepaid credit bundle
GCP project (Vertex TTS)	Usage	Folded into TTS	Service account for Gemini Cloud TTS
Sarvam / Gemini accounts	Setup	₹0 to start	Sarvam includes ₹1,000 free credits
Development & tuning	One-time	Delivered	Dialect prompt, lexicon, latency + filler tuning

8 Assumptions, notes & sources

Assumptions: balanced minute (caller ~30 s / bot ~30 s, ~4 turns); reply capped at 160 output tokens; ~5K-token dialect prompt in context; Gemini audio at 25 tokens/s; FX \$1 ≈ ₹86; Exotel outbound to Indian mobiles at ₹0.80–1.00/min + 18% GST. Actual cost varies with call length, talk-ratio, cache hit-rate, and your negotiated telephony rate.

Model note: the deployed LLM (gemini-3.1-flash-lite) and Gemini TTS (gemini-2.5-flash-tts) are priced at Google's published Flash-Lite / Flash-TTS tiers. Live-audio pricing applies only to the **gemini-direct** engine.

Sources (verified Jul 2026):

- Sarvam AI pricing — docs.sarvam.ai/api-reference-docs/pricing
- Gemini API pricing — ai.google.dev/gemini-api/docs/pricing
- Gemini audio tokenisation (25 tok/s) — Google AI Developers forum / docs
- Exotel pricing — exotel.com/pricing (credit-based; contact for exact per-min rate)

artintele.ai — Bengaluru-Kannada AI voice tele-caller · Sarvam × Gemini × Exotel · Pipecat orchestration. This brochure is indicative and for planning; list prices change — reconfirm at the sources above before contracting.